UPSC Prelims

Science & Tech

Class 1

Orientation, Biotechnology Basics

First Class : Science and Technology for UPSC Prelims 2023

- Orientation : Basic details about the course
- What you can expect from me
- What I expect from you
- Doubts / Queries and Everything else
- Academic : Basics of Biotechnology
- Some PYQs

Orientation : Teaching methodology and notes

- Teaching methodology : Slides on the digital board
- Static First, for conceptual clarity.
- Current Affairs integration
- PYQs : 2013 to 2022
- Notes : Slides will be shared, without annotation
- One Pager Notes at the end of the class
- Both of these can be downloaded from the portal.

Orientation : Class Schedule

Date	Day	Subject
15th March	Wednesday	Orientation and Biotechnology Basics
16th March	Thursday	Basic Biology: Cells and Genetics
17th March	Friday	Principles of Biotech and their applications
18th March	Saturday	Biotechnology Current Affairs
20th March	Monday	Health Issues
21st March	Tuesday	Nanotechnology
22nd March	Wednesday	Space Technology
23rd March	Thursday	Space and Défense 1
24th March	Friday	Défense 2
25th March	Saturday	Emerging Technology and Computers 1
27th March	Monday	Emerging Tech and Computers 2
28th March	Tuesday	Energy : Basics
29th March	Wednesday	IPR and Role of Scientists
30th March	Thursday	Reserve Class
31st March	Friday	Reserve Class

Orientation : Class timing

- Classes will be held from Monday to Saturday.
- No Class on Sunday : Your time to revise
- Class timing : **12 NOON**

Orientation : Doubts and Queries

- Answering doubts is extremely important.
- Completing the course on time is also very important.
- In the live class, I will ask you to participate. Put all your comments / queries / suggestions in the discussions board below. Whatever I can address during the class, I will.

Orientation : Doubts and Queries

- Remaining questions will be dealt with on the discussion board, and you will get the answers to your questions.

What you can expect from me

- Firstly, Thank you for placing your trust in me. I will give my best here.
- Coverage of all the static themes in science and technology.
- Barring any force majeure, the course will be finished on time and as per schedule.

Topics To Be Covered

- Biotechnology
- Health Issues
- Nanotechnology
- Space Technology
- Defence Sector
- Computers and Information Technology
- Energy
- Intellectual Property Rights
- Role of Indian Scientists

What you can expect from me

- Coverage of All Current Affairs : You DO NOT NEED TO READ PT 365 : I will cover the PT-365 of past 2 years.
- Coverage of ALL PYQs from 2013 to 2022
- Relevant Content : No mushrooming of Content. Only what is relevant. Our target here is to get a good percentage of the questions right.
- Our goal is not to become a scientist, but to get 100+ marks in prelims.

Utilizing this course effectively

- 15th March to 31st March : Science and Technology
- 3rd April to 20th April : Environment

March 2023

Sun	Mon	Tue	Wed	Thu	Fri	Sat
			1	2	3	4
5	6	7	8	9	10	11
12	13	14	15	16	17	18
19	20	21	22	23	24	25
26	27	28	29	30	31	

April 2023

Sun	Mon	Tue	Wed	Thu	Fri	Sat
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30						

Utilizing this course effectively

- Attend the courses live
- Spend 2-3 Hours here, after that DO NOT take a break.
- Spend at least 15 minutes revising what we have learnt from the slides and one pager notes.
- Next morning : Revise the previous session before coming here.
- Out of your 8 hours of daily study time, I'm asking for 4 hours.

What I expect from you

- Work hard and listen effectively.
- Use Active Recall to Revise.
- This is what your ideal schedule looks like :
- Revise yesterday's class in the morning
- 12pm onwards : Attend the class here
- After the class : Revise from the one pager for 15-20 minutes.
- Post lunch : Study a different subject / give test
- Sunday : Revise whatever has been taught the entire week.

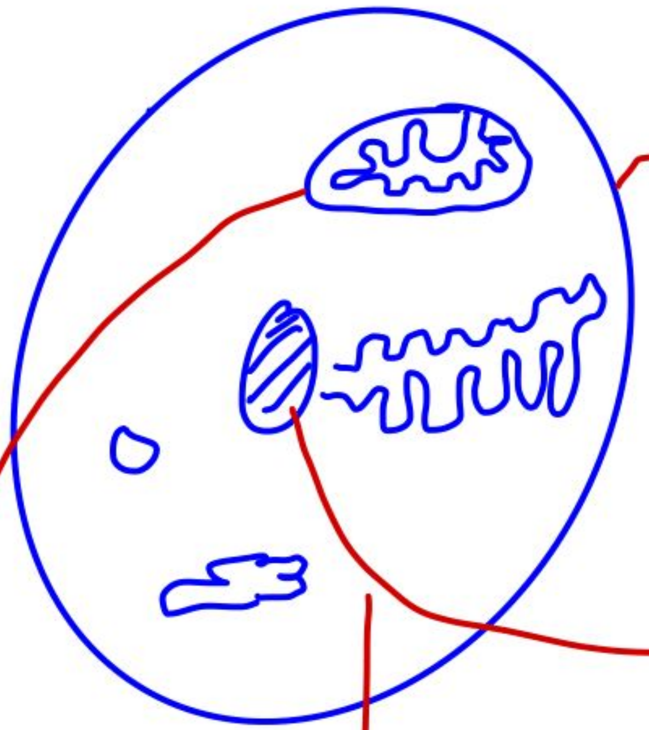
What I expect from you

- Live attendance of class Regularly
- Proper timely revision, daily and on Sunday
- Result : From today till 20th April, you will be done with Science and tech and Environment. Your 2 biggest fears will be sorted.
- If you follow the plan, you will have a set of one pagers which you will have revised from.
- 50-60 pages with relevant content that you can effectively revise in one day.

LET'S GET STARTED!

- First topic to cover : Biotechnology Basics.
- Some of this was covered in the demo class, but I will start assuming that you have not seen any of it.
- So if you have not read any science/ environment before today, don't worry. Leave it to me.
- Just stay attentive.
- First we will study biotechnology.

Basics of Biotechnology.



cell Membrane

cell: the basic unit of life [Fundamental]

Nucleus.

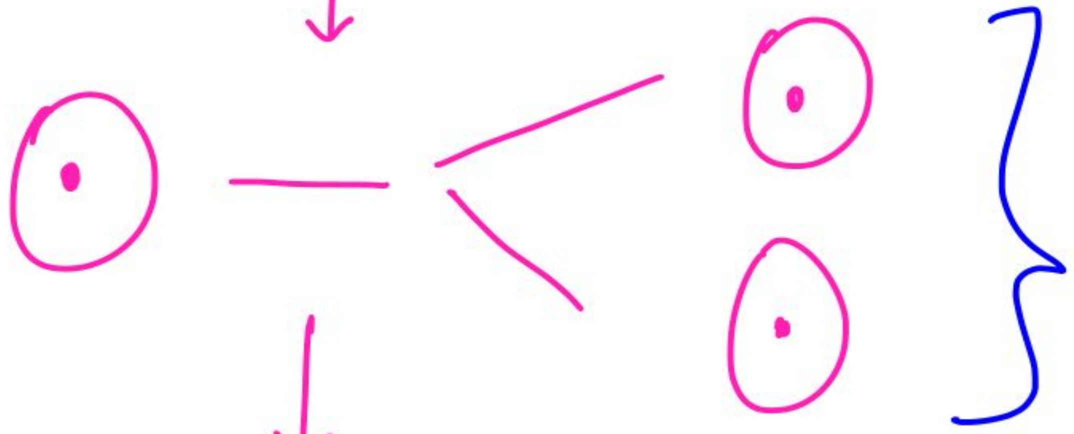
→ Brain of the cell.
→ controls cellular functions.

cytoplasm

cell organelles: ex: Mitochondria (powerhouse)

Growth

cell has to reproduce



information for transmission has to be stored somewhere.

growth can occur only when cells reproduce.

↓

important information to run the cell has to go from one cell to the next.

↓

This information needs to be stored somewhere.

'information' → DNA

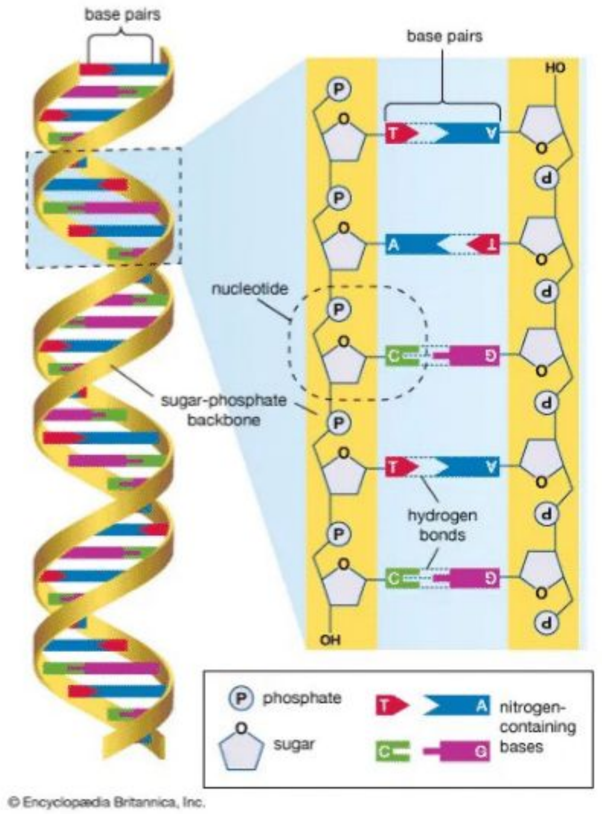
Nucleus

(deoxyribonucleic acid)

Ribose   acidic

DNA looks like:

double helix structure.



└→ zoom out →  → problem: very long (cell is small)

↑

every cell has complete DNA → Forensics

↑

solution

↑

Need to condense

↑

Lets see the condensation

Clarity for UPSC by Dr. Shivin

Ed Sarrthi Your Own Mentor

What do we know so far?

cell has to reproduce.

For reproduction, information has to pass to daughter cells.

DNA is the information inside a cell.

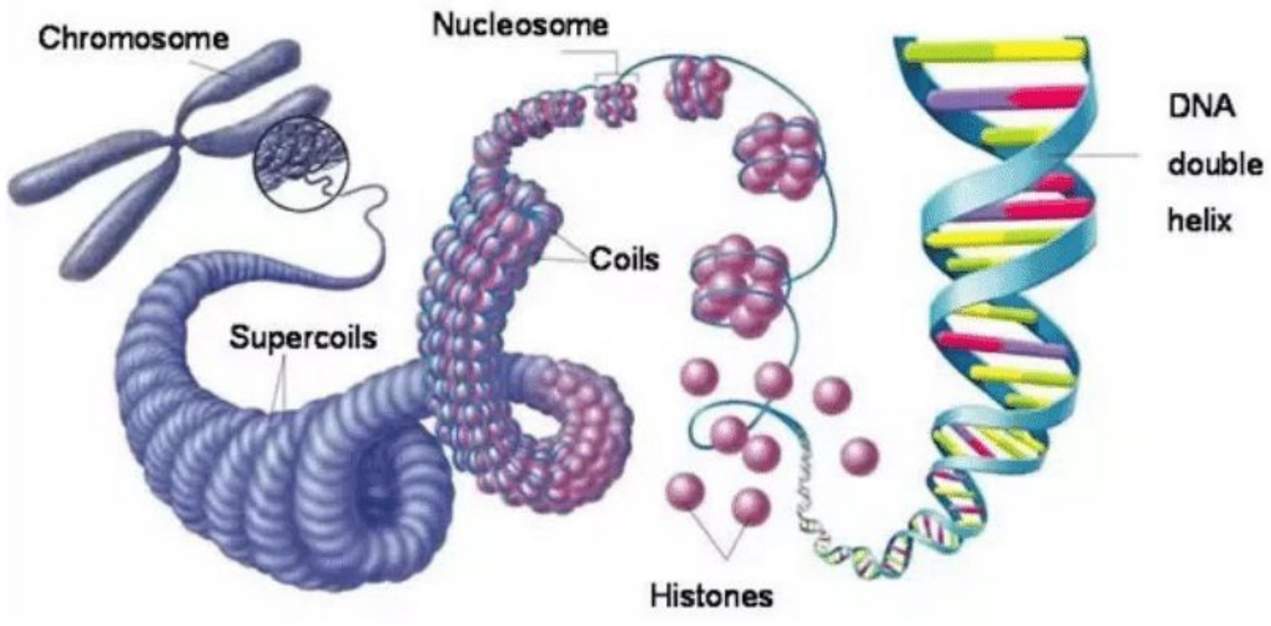
every cell has DNA

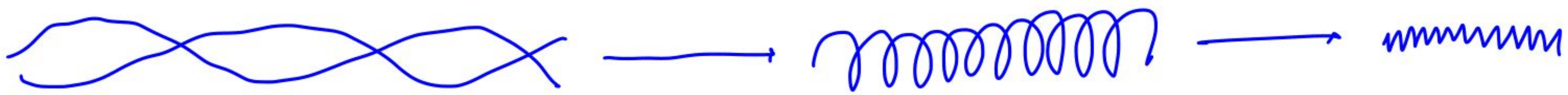
DNA is present in the NUCLEUS.

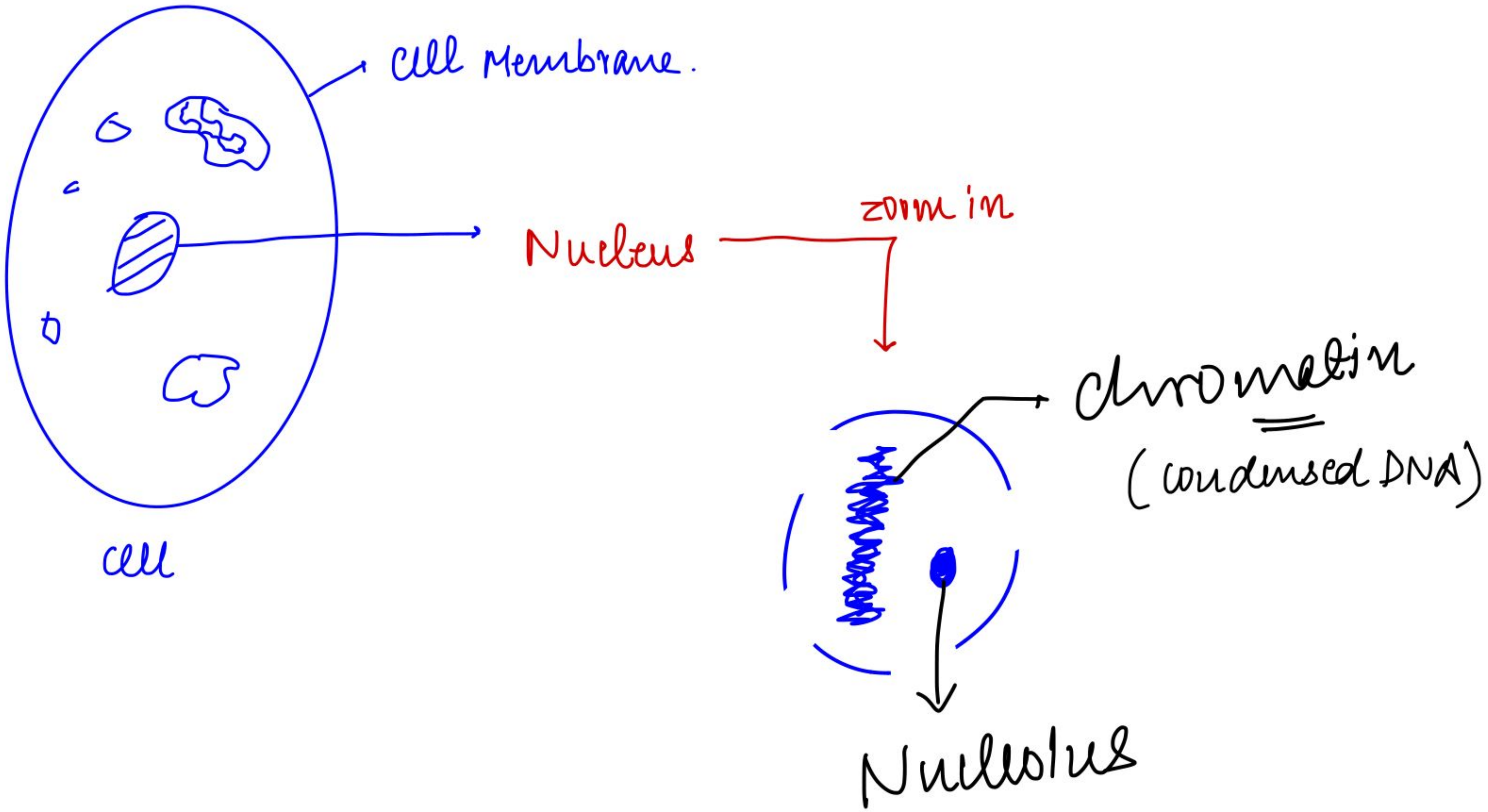
DNA has to be condensed as it is very long.

DNA is a double helix.

$\text{DNA} \xrightarrow[\text{Histones}]{\text{condensation}} \text{Chromatin} \xrightarrow[\text{condense}]{\text{further}} \text{Chromosomes.}$






cell Membrane.
Nucleus
zoom in
chromatin
(condensed DNA)
Nucleolus
cell

We Know: DNA is the information.

⇓

But WHY?

DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ proteins (enzymes)

↓

Ribonucleic acid.

2 Questions

- why RNA?
- why protein?

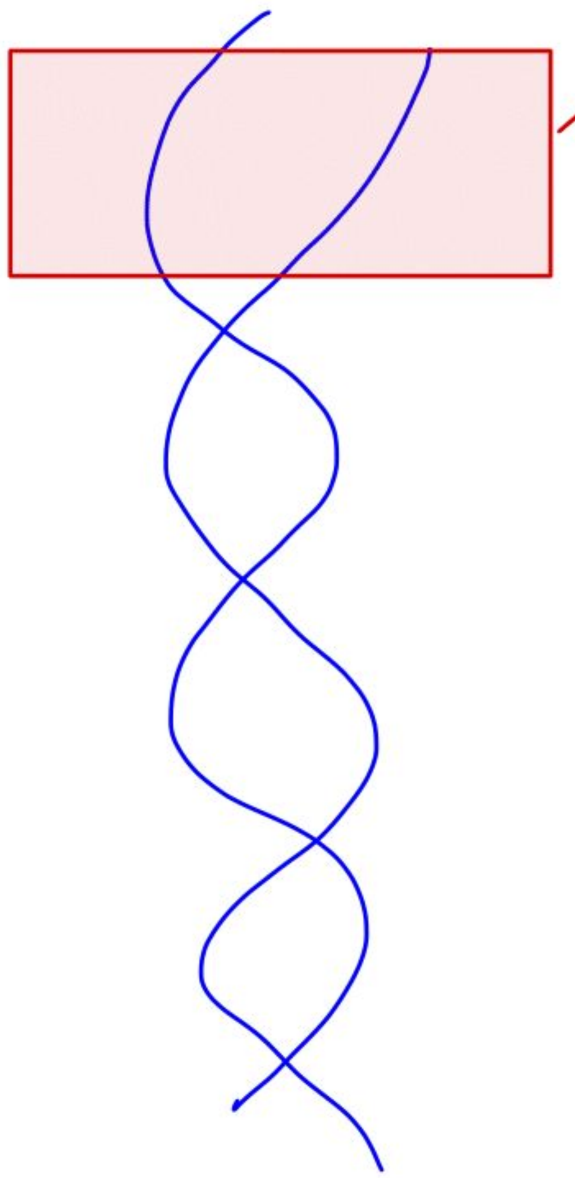
Why proteins (enzymes)?

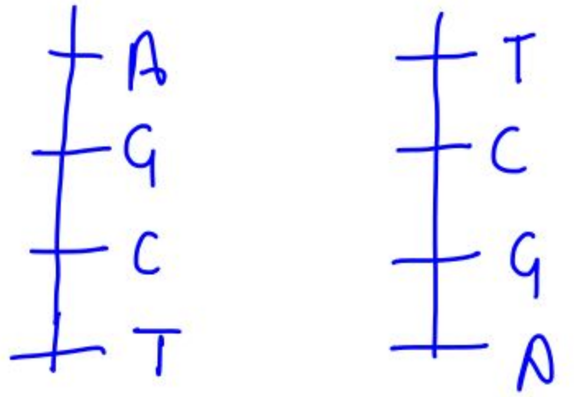
Sucrose (table sugar) $\xrightarrow{\text{sucrase}}$ glucose + fructose $\xrightarrow{\text{enzymes}}$ energy

(Kreb's cycle)

For a cell to reproduce → New cell has to form → New cell must have all essential enzymes

enzymes come from DNA.





Adenine (A)
Thymine (T)

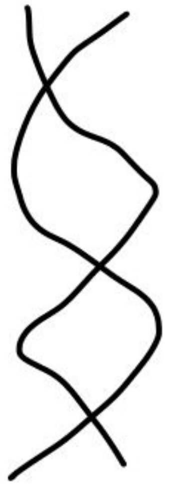
Guanine (G)
Cytosine (C)

At the girl's club

double Helix → difficult to process → make it into single strand → RNA

RNA ↓ proteins (enzymes)

DNA v/s RNA



double Helix

single strand.

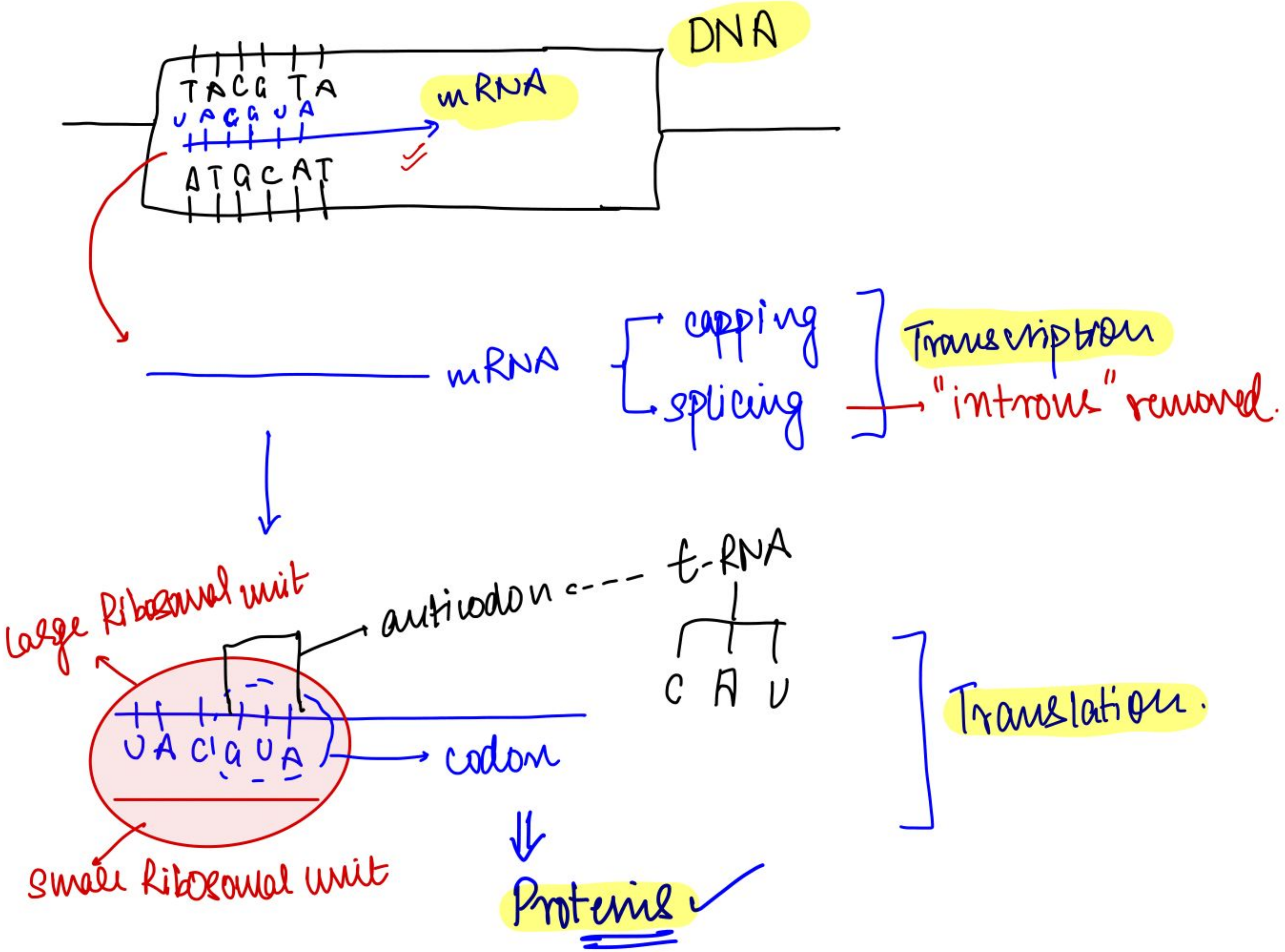
A | T | G | C

A | U | G | C

$DNA \xrightarrow{\text{transcription}} RNA$

$RNA \xrightarrow{\text{translation}} \text{proteins}$

U → Uracil.


DNA
TACG TA
UACGUA
mRNA
ATGCAT
mRNA
capping
splicing
Transcription
"introns" removed.
t-RNA
anticodon
CAU
Large Ribosomal unit
UAC GUA
codon
Translation.
Small Ribosomal unit
Proteins

What do we know?

cell has to reproduce.

For reproduction, information has to pass to daughter cells.

DNA is the information inside a cell.

every cell has DNA

$$\underbrace{\text{DNA} \xrightarrow{\text{transcription}} \text{RNA} \xrightarrow{\text{translation}} \text{protein (enzymes)}}_{\text{central law of genetics : Living cells.}}$$

If there is a law, there is an exception

article 14 → Right to Equality ⟶ exception: No arrest of President.

Similarly

Law of DNA ⟶ RNA ⟶ protein [Has an exception]

⇓

exception

↓

Some viruses

Recap

'life' → Basic unit → cell.

∴ if you dont have cell → you are non-Living

Feature: Reproduction (Need of DNA, RNA, protein)

⇒ only Living can reproduce through DNA, RNA, protein.

↓

virus is an exception

Viruses.

- cell — Basic unit of life.
- virus — Borderline between living and non living.

why?

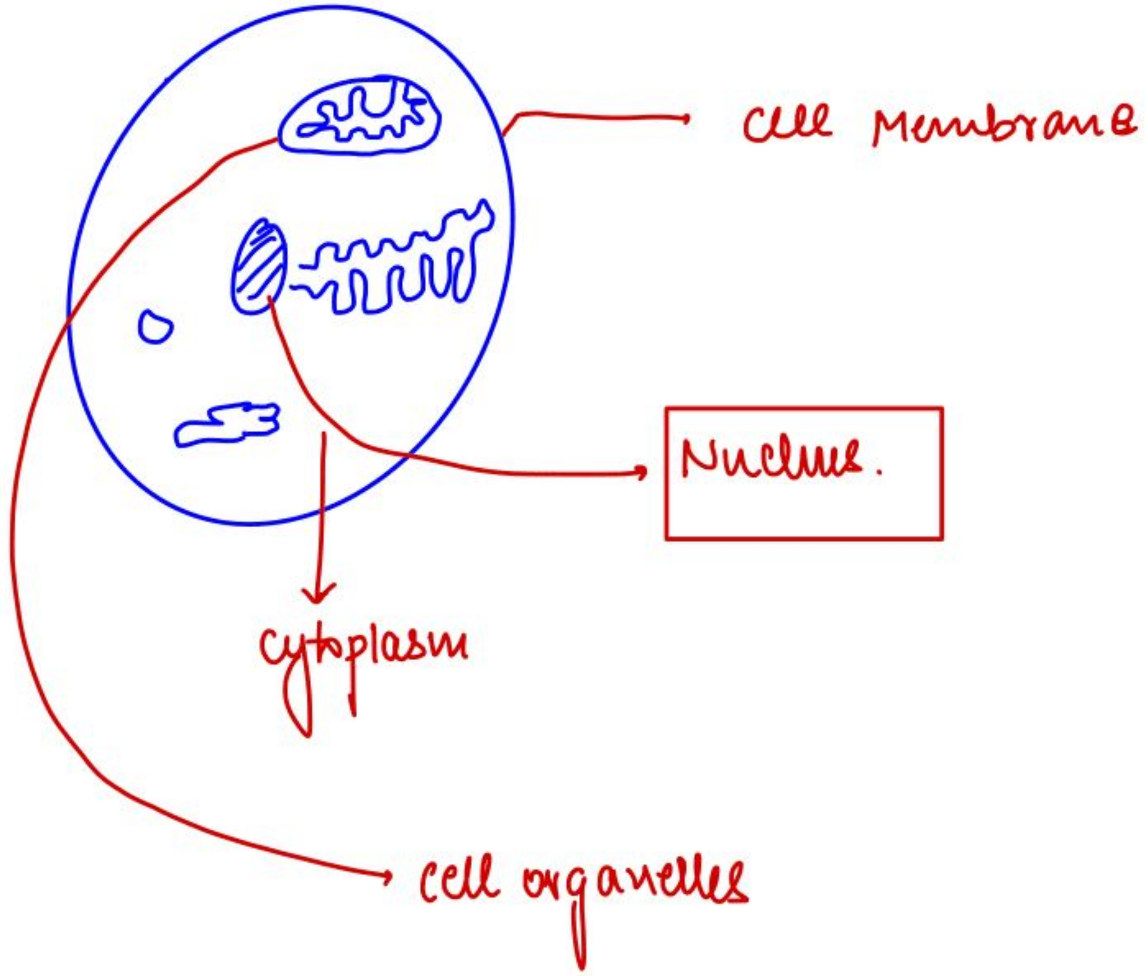
- viruses have DNA or RNA and some proteins
- They can reproduce in host cell.

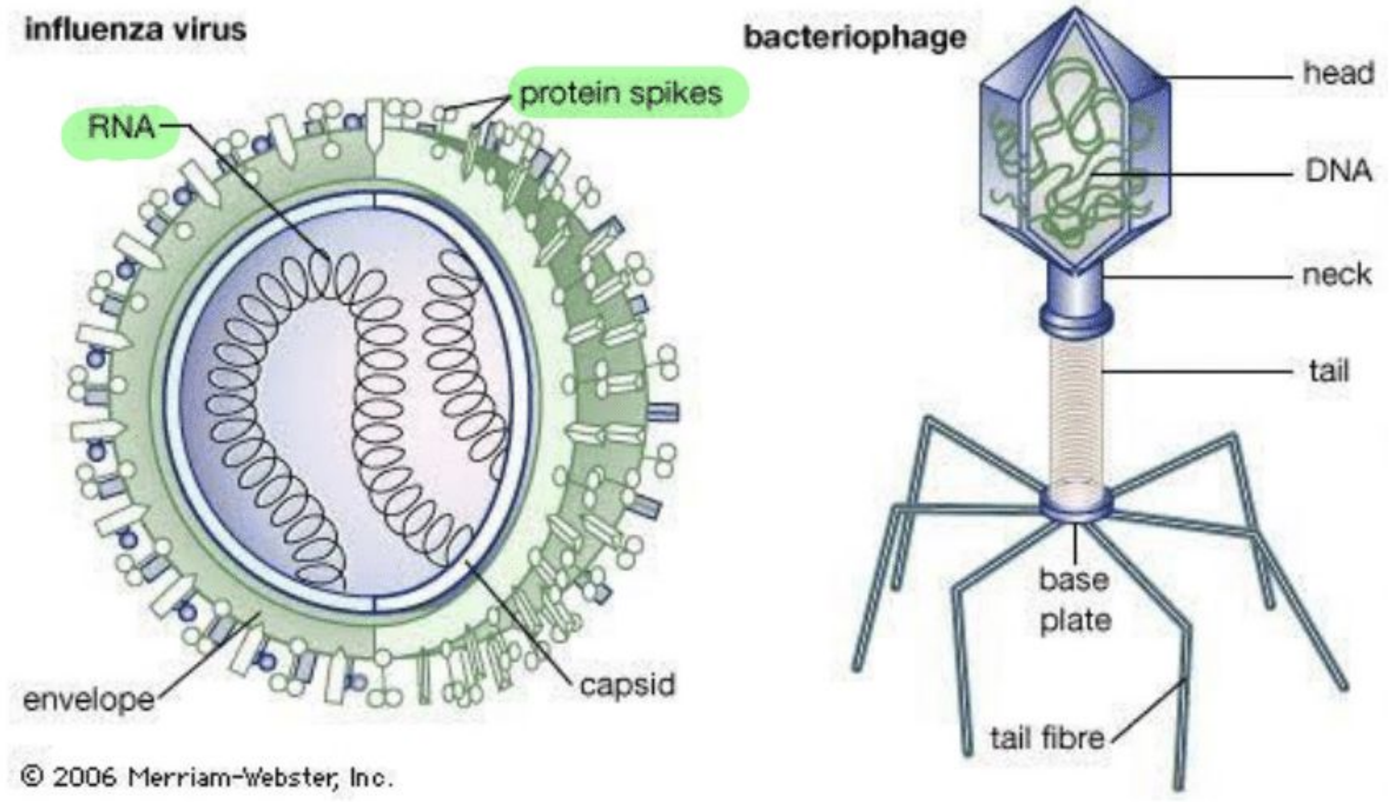
They dont have their own cellular structure.

Let's compare.

cell

virus.





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Features of a virus.

1. Has DNA, RNA, protein
2. Can Reproduce, inside a host cell.
3. No cellular structure.

Borderline.

How do they reproduce inside host cell?

L think

if you cannot.....

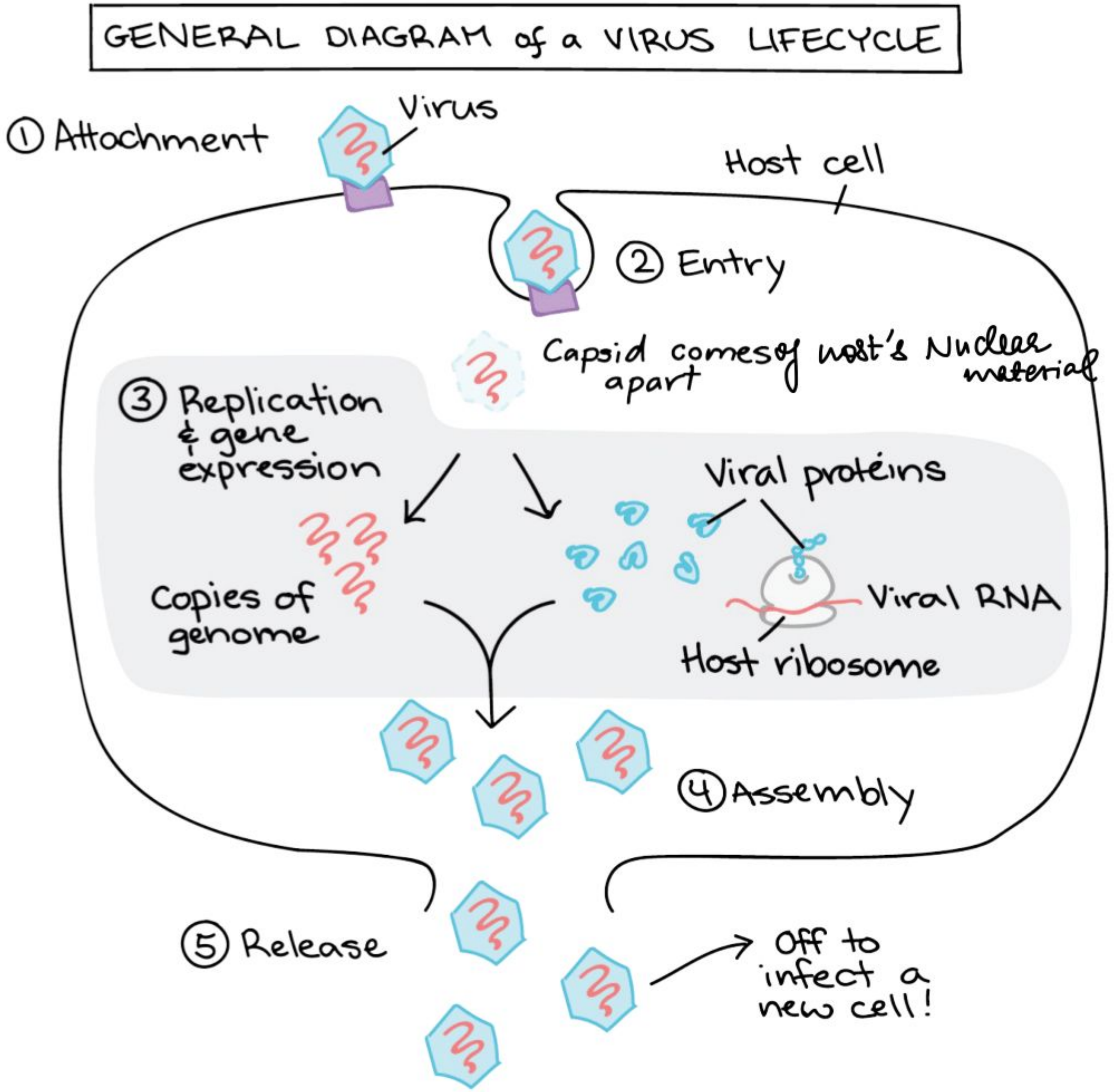
you ask your friend.

virus → uses DNA, RNA (called genetic material) of host
↓
integrates with it to reproduce (Multiply)

virus multiplies
↓
Host dies
(ex: Covid 19)

Host? → cells that virus infects.

covid-19 virus → Host : Humans.



original | infected
DNA ← - - viral DNA
↓
RNA ← - - - virus RNA
↓
protein ← - - virus proteins

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Original Process

DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ Protein

Virus: Starts with RNA → But needs to integrate with DNA so that it can use Host cell mechanism

Remember:

DNA v/s RNA

double Helix

single strand.

DNA: A | T | G | C

RNA: A | U | G | C

DNA $\xrightarrow{\text{transcription}}$ RNA

RNA $\xrightarrow{\text{translation}}$ protein

U → Uracil.

What does virus need to do?

↓

think

(diesel v/s petrol)

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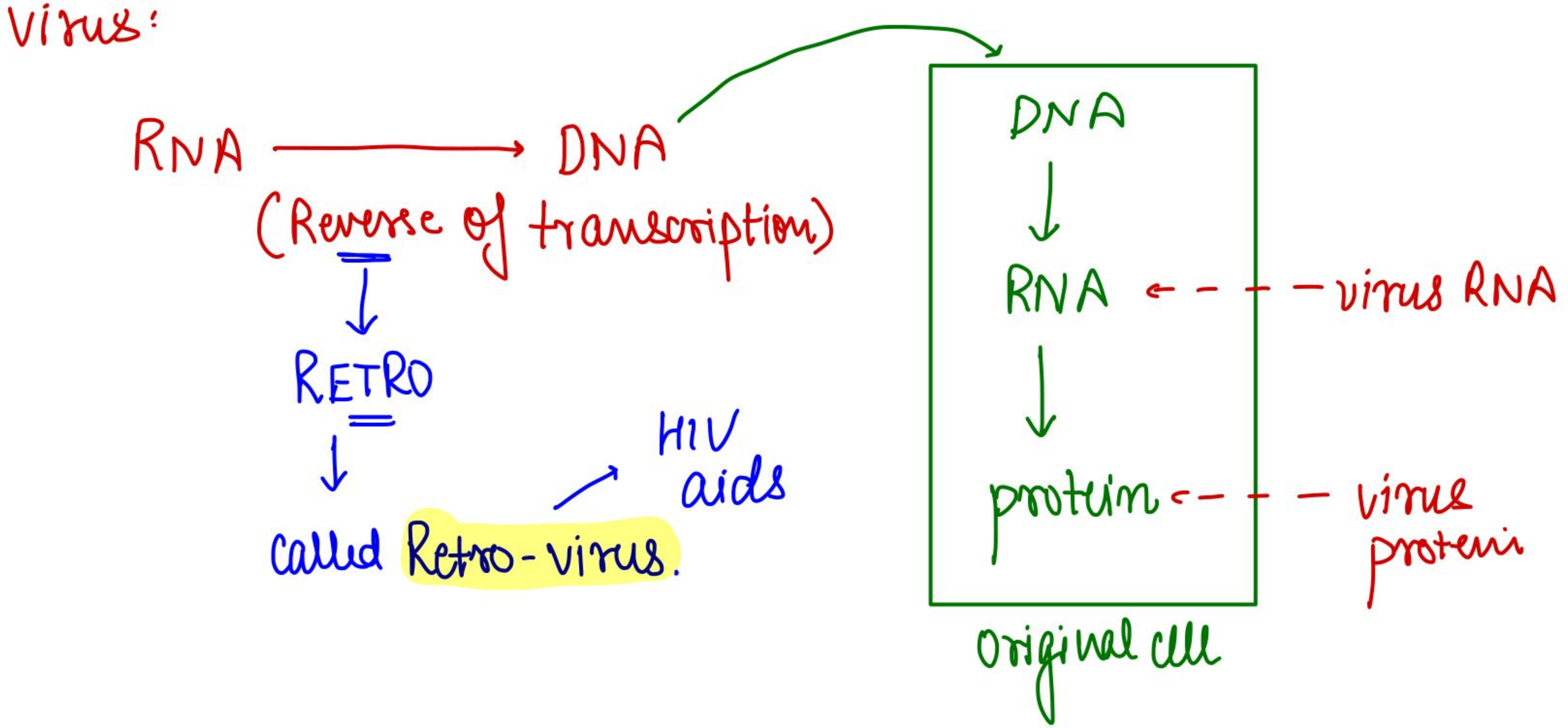
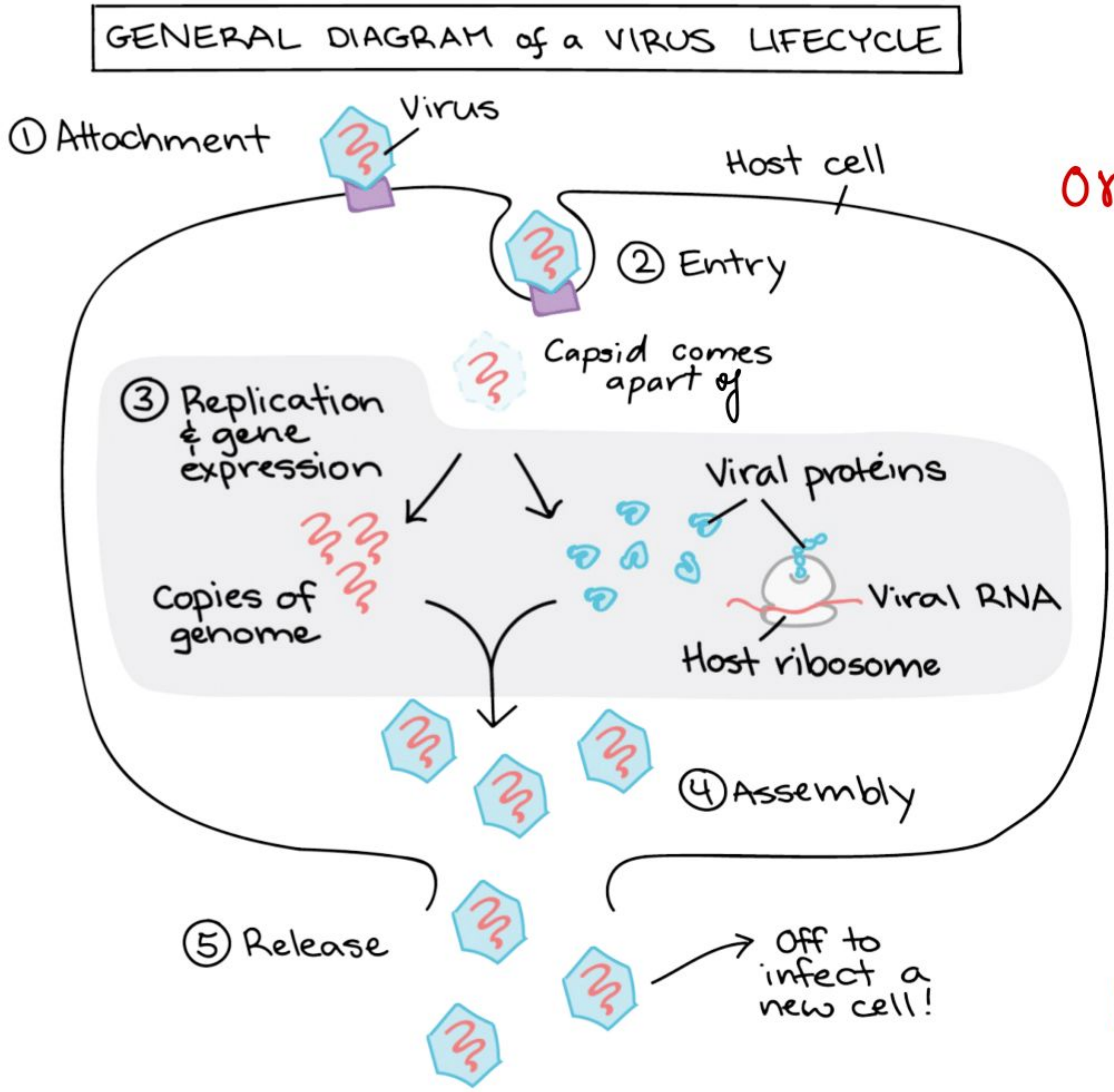
- virus needs to convert its own RNA → DNA
- then virus can easily integrate c̄ host cell.

Virus:





original virus:
RNA + protein
↓ through HOST cell
Many RNA + Many protein
↓
Many virus.

DNA $\xrightarrow{\text{transcription}}$ RNA $\xrightarrow{\text{translation}}$ protein] Normal (central dogma)

virus

RNA → DNA (integrates with host)

↓

Reverse of original : $\boxed{\text{Retro}}$ - virus

ex : HIV (AIDS)

assembly of new virus.

⇓

RNA + protein

⊛ Remember: virus can do this only inside HOST, NOT independently.

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Now, I will show you a video of HIV infecting a Cell

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HIV

- infects CD4+ T Lymphocyte.
- CCR5 / CXCR4 Receptor
- Enters the cell
  - ↳ 3 enzymes → Reverse Transcriptase
  - → Integrase
  - → Nuclease.
- we can block it at any stage to prevent HIV infection.

Let’s Solve Some Previous Year Questions!

Previous Year Questions: 2016

Q. In the context of the developments in bioinformatics, the term ‘transcriptome’, sometimes seen in the news, refer to -

a) a range of enzymes used in genome editing
b) the full range of mRNA molecules expressed by an organism
c) the description of the mechanism of gene expression
d) A mechanism of genetic mutation taking place in cells

Answer: B

- The transcriptome is the set of all RNA molecules in one cell or a population of cells.
- It is sometimes used to refer to all RNAs, or just mRNA, depending on the particular experiment.
- It differs from the exome in that it includes only those RNA molecules found in a specified cell population, and usually includes the amount or concentration of each
- RNA molecule in addition to the molecular identities

Previous Year Questions: 2016

Q.Which of the following statements is/are correct?

Viruses can infect -

1. Bacteria
2. Fungi
3. Plants

Select the correct answer using the code given below:

a) 1 and 2 only

b) 3 only

c) 1 and 3 only

d) 1, 2 and 3

Answer: D

- A virus is a small infectious agent that replicates only inside the living cells of other organisms. Viruses can infect all types of life forms, from animals and plants to microorganisms, including bacteria and fungi.

- According to the type of the host they infect, viruses are classified mainly into the following four types:
  a. Plant viruses including algal viruses-RNA DNA
  b. Animal viruses including human viruses- DNA/RNA
  c. Fungal viruses(Mycoviruses)-ds RNA
  d. Bacterial viruses (Bacteriophages) including cyanophages-DNA

Previous Year Questions: 2021

Q.Consider the following:

1. Bacteria
2. Fungi
3. Virus

Which of the above can be cultured in an artificial/synthetic medium?

a) 1 and 2 only

b) 2 and 3 only

c) 1 and 3 only

d) 1, 2 and 3

Answer: A

- Bacteria and Fungi can be cultured in an artificial/synthetic medium. Whereas viruses require a living host cell for replication. Infected host cells (eukaryotic or prokaryotic) can be cultured and grown, and then the growth medium can be harvested as a source of the virus. Hence, 1 and 2 are correct and 3 is not correct.

- Therefore, option (a) is the correct answer

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Previous Year Questions: 2013

Q.Which of the following statements is/are correct?

1. Viruses lack enzymes necessary for the generation of energy.
2. Viruses can be cultured in any synthetic medium.
3. Viruses are transmitted from one organism to another by biological vectors only.

Select the correct answer using the codes given below

a) 1 only
b) 1 and 2 only
c) 1 and 3 only
d) 1, 2 and 3

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Answer: A

- Since Viruses lack organelles and are totally dependent on the host cells metabolic machinery for replication, (hence) they cannot be grown in synthetic media.
- In the laboratory, animal viruses are grown in animals or animal cell culture and plant viruses are grown in plants or plant cell culture.
- There are two categories of Vectors: biological and mechanical.
- As was stated earlier, if the virus increases its number while in association with a vector, then that vector is termed as being biological.
- Conversely, the vector is termed to be mechanical, if the virus doesn't increase its number while in association with that vector.
- Arthropods such as wasps, which repeatedly can sting multiple animals, could serve as mechanical vectors by transporting viruses on the surface of their stingers.

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Previous Year Questions: 2021

Q. Consider the following statements:

1. Adenoviruses have single-stranded DNA genomes whereas retroviruses have double-stranded DNA genomes.
2. Common cold is sometimes caused by an adenovirus whereas AIDS is caused by a retrovirus.

Which of the statements given above is/are correct?

a) 1 only

b) 2 only

c) Both 1 and 2

d) Neither 1 nor 2

Answer: B

- Adenovirus is a type of virus that has no envelope whereas retroviruses are characterized as enveloped viruses. Adenoviruses have double-stranded linear DNA and are associated with two major core proteins. A retrovirus is a virus that uses RNA as its genetic material. When a retrovirus infects a cell, it makes a DNA copy of its genome that is inserted into the DNA of the host cell. Hence, statement 1 is not correct.

- Adenoviruses are common viruses that cause a range of illnesses. They can cause cold-like symptoms, fever, sore throat, bronchitis, pneumonia, diarrhoea, and pink eye (conjunctivitis). Whereas, retroviruses can cause several human diseases such as some forms of cancer and AIDS. Hence, statement 2 is correct.

- Therefore, option (b) is the correct answer.

Previous Year Questions: 2022

Q. Consider the following statements: DNA Barcoding can be a tool to:

1. assess the age of a plant or animal.
2. distinguish among species that look alike.
3. Identify undesirable animal or plant materials in processed foods.

Which of the statements given above is/are correct?

(a) 1 only

(b) 3 only

(c) 1 and 2

(d) 2 and 3

Answer: D

- DNA barcoding is a system for species identification focused on the use of a short, standardized genetic region acting as a “barcode”
- DNA barcoding has got no association with determining the age of an organism.
- DNA barcoding is a system for fast and accurate species identification that makes ecological system more accessible by using short DNA sequences instead of whole genome and is used for eukaryotes.
- In recent times, DNA barcoding has emerged as the most potent tool for detection of adulteration of food with unwanted plant and animal material.

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Basics of Biotechnology

- **Cell**
  - Basic unit of life
  - Has Nucleus, Cell membrane, cytoplasm and Cell organelles.
  - For growth and reproduction, important information needs to be transmitted.
  - Information is DeoxyRiboNucleic Acid (DNA)
  - Long DNA molecule needs to be condensed.
  - DNA to Chromatin to Chromosomes
- **DNA to RNA to Protein (Central Dogma)**
  - Transcription : DNA to mRNA
  - Translation : mRNA to Protein, tRNA and rRNA help in this process
  - Why RNA : Processing is easier
  - RNA vs DNA
    - Single Strand
    - Uracil in place of thymine
  - Why Protein : Enzymes for cellular function
- **Virus**
  - Borderline between living and non living
  - DNA / RNA and Protein but no cellular machinery
  - Can reproduce only inside host cell by integrating its genetic material
  - Some viruses start with RNA : Retro Virus
  - Reverse Transcription by Retro Virus
- **PYQs**
  - Can virus be cultured in artificial medium?
  - What genome does Retro virus have? DNA or RNA?
  - What is a transcriptome?
  - DNA Barcoding : can it be used to asses age?

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